

Evaluation Guide

Lab Report · Project Proposal · Final Presentation

HW/SW Co-Design 2025/26 · Instituto Superior Técnico

Three Assessed Deliverables

Lab Report	Project Proposal	Final Presentation
GEMM Implementation	Week 4 Presentations	Week 7 Presentations
▶ SW Float baseline ▶ SW Fixed-point + error analysis ▶ HW/SW Float: AXI-Lite + AXI-Stream/DMA ▶ HW/SW Fixed-point IP ▶ Conclusion & trade-offs 20 pts Due: Week 7	▶ Problem statement & motivation ▶ Algorithm + fixed-point considerations ▶ HW/SW partitioning rationale ▶ HLS plan, Vivado integration ▶ Expected performance & timeline 20 pts 15 slides	▶ Implemented architecture & HLS results ▶ System integration + ILA debugging ▶ SW application + timing measurement ▶ Measured speedup + resource usage ▶ Challenges & lessons learned 20 pts 15 slides

Submission (Week 7): both projects are due at the same deadline. Submit a single zip file containing one folder per project (*matrix_multiply/* and *final_project/*); each folder must include the code and the lab report following the given template. The Project Proposal presentation remains a separate deliverable assessed in

Topics Coverage — Are You Being Exhaustive?

Each topic below must be covered **at least once** across your two projects — either in the **Matrix Multiply** lab report or in the **Final Project**. You must include a **topic coverage matrix** in both your final report and your final presentation, indicating in which project each topic was addressed.

Topic	Description
AXI-Lite	Register-based control interface
AXI-Stream	High-throughput streaming
Float × Fixed	Data-type trade-off in SW and HW
ILA Debugging	In-system probing, Vivado HW Manager
BRAM Storage	Local buffers in HLS IP
Cache Flush/Inv.	Xil_DCacheFlushRange for DMA coherency
HLS Pragmas	PIPELINE, UNROLL, ARRAY_PARTITION
DMA Config.	AXI DMA IP, HP port, XAxiDma_* driver
C Testbench	Golden reference, tolerance, co-sim waveforms
XTime Timing	Timers outside printf; COUNTS_PER_SECOND
HW/SW Partition	Amdahl analysis; which tasks benefit from PL

PART A

Lab Report

HW/SW GEMM Implementation — 20 pts

Lab Report — Score Distribution

Sec 1: Introduction		2
Sec 2: SW Float		3
Sec 3: SW Fixed-Point		3
Sec 4: HW/SW Float		5
Sec 5: HW/SW Fixed-Point		5
Sec 6: Conclusion		2

- ▶ **Sec 1 — Introduction:** Title page · Motivation · Lab objectives
- ▶ **Sec 2 — SW Float:** Algorithm correctness · Code quality · Execution time vs. N
- ▶ **Sec 3 — SW Fixed-Point:** Q-format choice · Timing comparison · MSE error analysis
- ▶ **Sec 4 — HW/SW Float:** AXI-Lite + AXI-Stream/DMA integration · Resource report · Speedup
- ▶ **Sec 5 — HW/SW Fixed-Point:** ap_fixed IP · Float vs. Fixed accuracy trade-off
- ▶ **Sec 6 — Conclusion:** Trade-off discussion · Lessons learned

⇒ **All developed code must be submitted together with the report so that the claims can be verified.**

Section 1 — Introduction

Sub-item / Criterion	What we assess	Pts
1.1 Motivation	Need for hardware acceleration explained; Zybo-Z7 / Zynq-7000 architecture benefits stated.	1
1.2 Objectives	Primary goals listed (float/fixed implementations, AXI-Lite vs. AXI-Stream/DMA comparison, speedup targets).	1
1.3 Report Structure	Brief overview of each subsequent section provided. <i>Penalty if absent.</i>	-1

Section 2 — Software Implementation: Float

Sub-item / Criterion	What we assess	Pts
2.1 Algorithm Description	Triple-nested loop shown; $O(n^3)$ complexity stated; float/double type justified; variable naming clear.	1
2.2.1 Code Size	.text / .data / .bss from compiler size output; heap and stack sizes stated; total in hex or decimal.	1
2.2.2 Execution Time vs. Matrix Size	≥ 4 matrix sizes tested (e.g. 64–1024); table + graph; $O(n^3)$ growth discussed; XTime_GetTime used (excludes I/O).	1

Section 3 — Software Implementation: Fixed-Point

Sub-item / Criterion	What we assess	Pts
3.1 Algorithm Changes & Q-Format	Q-format selected with justification; integer arithmetic shown; overflow strategy described (saturate/scale). <i>Penalty if absent.</i>	-1
3.2.1 Code Size	.text / .data / .bss reported; heap and stack sizes stated; size difference vs. float version noted and explained.	1
3.2.2 Execution Time vs. Matrix Size	Table + graph; comparison with SW float baseline; speedup factor computed.	1
3.2.3 Error Analysis vs. Float Baseline	MSE computed per matrix size; precision loss discussed; ≥ 2 Q-formats compared; pass/fail criterion defined.	1

Sub-item / Criterion		What we assess	Pts
4.1	AXI-Lite IP	PIPELINE / UNROLL pragmas applied and shown; register map documented; C-sim PASS; co-sim waveforms included.	1
4.2	AXI-Stream DMA IP	+ hls::stream interface described; XAxiDma configured; cache Flush/Invalidate applied for DMA coherency; ILA waveforms included.	1

Section 4 — HW/SW Float: Results

Sub-item / Criterion	What we assess	Pts
4.3.1 Code Size	SW driver .text / .data / .bss reported for both AXI-Lite and AXI-Stream/DMA versions; heap and stack sizes stated.	0.75
4.3.2 Hardware Resource Usage	Post-synthesis: LUT/FF/DSP/BRAM for both IPs; ≥ 100 MHz confirmed; BRAM for local buffers identified.	0.75
4.3.3 Execution Time vs. Matrix Size	Both IPs timed using XTime_GetTime; total HW/SW time reported per matrix size; table + graph vs. SW baseline.	0.75
4.3.4 Speedup Comparison	$S = T_{\text{sw}}/T_{\text{hw}}$ per matrix size; bar/line chart; bottleneck identified; speedup scaling with N discussed.	0.75

Section 5 — HW/SW Fixed-Point

Sub-item / Criterion	What we assess	Pts
5.1 Implementation Details	ap_fixed<W,I> types for inputs/outputs/accumulators with justification; BRAM for local buffers; PIPELINE II=1 achieved; quantisation mode (AP_RND/AP_SAT) selected.	2
5.2.1 Code Size	.text / .data / .bss reported; heap and stack sizes stated; difference from float version noted.	0.75
5.2.2 Hardware Resource Usage	LUT/DSP/BRAM vs. float IP — table + commentary; fixed-point advantage in resource count demonstrated.	0.75
5.2.3 Execution Time vs. Matrix Size	Table + graph; comparison with float HW/SW; speedup or parity explained.	0.75
5.2.4 Float vs. Fixed Comparison	Side-by-side: resources, accuracy (MSE), execution time; justified recommendation; ≥ 1 comparison chart.	0.75

Section 6 — Conclusion

Sub-item / Criterion	What we assess	Pts
Summary of Findings	All four implementations addressed (SW float, SW fixed, HW float, HW fixed); key numbers cited.	1
Trade-Off Discussion	Precision vs. performance vs. resources articulated clearly; design recommendation with justification given.	1

PART B

Project Proposal

Week 4 Presentations — 20 pts

Project Proposal — Score Distribution

Problem Statement & Motivation		2
Algorithm Description		3
HW/SW Partitioning		3
Hardware Architecture		3
HLS Plan + Testbench		3
System Integration		2
Expected Performance		2
Timeline & Risks		1
Presentation Quality		1

Project Proposal — Evaluation Criteria (1 / 2)

Sub-item / Criterion	What we assess	Pts
Problem Statement & Motivation	Application described; bottleneck identified; motivation for HW acceleration convincing; Amdahl insight attempted.	2
Algorithm Description	Inputs/outputs/O(?) defined; pseudocode or flowchart present; parallelism identified; fixed-point considerations addressed (Q-format or N/A justified).	3
HW/SW Partitioning	PS/PL task split with rationale; AXI type selected and justified (Lite vs. Stream); data flow described; DMA considered if streaming.	3
Hardware Architecture	Block diagram present; AXI details (width, clock, II target); BRAM plan for local buffers; ap_fixed<W,I> or float stated; key HLS directives planned.	3
HLS Plan + Testbench	Specific directives listed (PIPELINE II=1, UNROLL, ARRAY_PARTITION); golden-reference testbench described; expected II/latency/resources estimated.	3

Project Proposal — Evaluation Criteria (2 / 2)

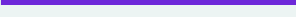
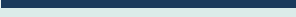
Sub-item / Criterion	What we assess	Pts
System Integration	Vivado block design topology described; address map planned; cache coherency considered (flush/invalidate); ILA monitoring plan included.	2
Expected Performance	SW baseline estimation method described; target speedup justified; DMA overhead acknowledged; XTime_GetTime methodology correct.	2
Timeline & Risks	Realistic week-by-week plan for Weeks 5–7; ≥ 2 project-specific risks with concrete mitigations.	1
Presentation Quality	Diagrams on key slides (architecture, partitioning); no placeholder text; template complete; delivered on time; Q&A handled.	1

PART C

Final Presentation

Week 7 Presentations — 20 pts

Final Presentation — Score Distribution

Overview & Objectives		1
Algo. & HW/SW Split		2
Hardware Architecture		2
HLS Implementation		3
System Integration		2
Software Application		2
Results		4
Challenges & Concl.		2
Presentation Quality		2

⇒ **All developed code and a written report (structured like the Lab Report) must be submitted together with the presentation. Both the code and the report must support and substantiate the claims made in the slides.**

Final Presentation — Evaluation Criteria (1 / 2)

Sub-item / Criterion	What we assess	Pts
Overview & Objectives	Application and objectives stated; comparison vs. proposal targets (met/not met); scope changes from proposal explained.	1
Algo. & HW/SW Split	Final PS/PL diagram shown; rationale for any changes from proposal; interface type and data flow confirmed.	2
Hardware Architecture	Implemented block diagram shown; IP ports/data width/clock stated; BRAM usage for local buffers explained; float vs. ap_fixed choice justified.	2
HLS Implementation	Directives with rationale shown; II achieved + schedule view discussed; C-sim & co-sim results stated; float vs. fixed decision justified.	3
System Integration	Vivado block design shown; ILA signals named and waveforms interpreted; cache flush/invalidate in SW explained; DMA setup described.	2

Final Presentation — Evaluation Criteria (2 / 2)

Sub-item / Criterion	What we assess	Pts
Software Application	Init → execute → verify flow shown; XTime_GetTime timing outside printf; COUNTS_PER_SECOND used; correctness check vs. SW reference shown.	2
Results	SW and HW timing measured; $S = T_{\text{sw}}/T_{\text{hw}}$ charted (≥ 3 sizes); resource table (LUT/FF/DSP/BRAM); DMA overhead isolated; meets/misses proposal target explained.	4
Challenges & Concl.	≥ 2 real technical challenges with root cause + fix (not just “Vivado crashed”); substantive lessons; concrete future work suggestions.	2
Presentation Quality	Diagrams on all architecture/result slides; no placeholder text; delivered within 15 min; clear Q&A answers.	2

Master Grading Summary

Lab Report	20	Project Proposal	20	Final Presentation	20
Introduction	2	Problem Statement	2	Overview & Objectives	1
SW Float	3	Algorithm Description	3	Algo. & HW/SW Split	2
SW Fixed-Point	3	HW/SW Partitioning	3	HW Architecture	2
HW/SW Float	5	HW Architecture	3	HLS Implementation	3
HW/SW Fixed	5	HLS Plan + Testbench	3	System Integration	2
Conclusion	2	System Integration	2	Software Application	2
		Expected Performance	2	Results	4
		Timeline & Risks	1	Challenges & Concl.	2
		Presentation Quality	1	Presentation Quality	2